MICAH BAXTER

mbbax@outlook.com • LinkedIn: Micah Baxter • Website: https://bax1906.github.io/micahbaxter/

EDUCATION

Bachelor of Science in Mechanical Engineering Expected May 2029

Liberty University, Lynchburg, VA

- Physics Minor, 3.87 GPA
- Organizations: Liberty Rocketry Competition Team
- Full Academic Scholarship, ACT: 35

TECHNICAL SKILLS

Software & Programming: SolidWorks, Onshape, Altium, KiCad, C#, C/C++, Python

Machines: Bambu Labs P1S combo (Bambu Studio), Creality Ender 3 (Cura), Shop Tools

EXPERIENCE

Chief Systems Engineer Present

Liberty Rocketry Competition Team, Lynchburg, VA

- Leading a team of 50+ engineers to develop an advanced high-power rocket for the IREC 2027 competition
- Coordinating between eight sub teams and multiple faculty advisors, providing technical guidance through the integration of the team's first active control system

Information Technology Intern Summer 2026

Medical Properties Trust, Birmingham, AL

- Resolved hardware and software issues including printer connectivity, email configuration, and network problems in a corporate healthcare environment
- Configured new employee workstations for onboarding, ensuring systems met security and access standards
- Assisted IT administrators in managing Active Directory including directory searches and file access permissions for employees

Avionics Design Engineer 2025-2026

Liberty Rocketry Competition Team, Lynchburg, VA

- Developed a custom 6-layer, STM32-based flight computer to control critical sensor and telemetry operations for a high-power competition rocket
- Appointed as Liberty Rocketry Chief Engineer for the upcoming competition season

PERSONAL PROJECTS

Thrust Vector Control Rocket 2023-2025

- Engineered an autonomous launchpad with launch clamps, a strongback, and flame trenches
- Designed and flew two TVC rockets across different airframe scales with a custom PCB flight computer and deployable catch points to enable future catch attempts, inspired by SpaceX's Starship

Embedded Systems and PCB Design 2023-2026

- Developed an STM32-based microcontroller ground computer for an upgraded launchpad that implements I2C, SPI, and UART communication and includes PWM controlled GPIO headers for modularity
- Designed and manufactured an STM32-based flight computer for a high-power collegiate rocket that gathers sensor and GPS data and transmits live telemetry for visualization on a custom ground station GUI

Puzzle Boxes 2025

- Invented and branded Baxter Boxes, a unique line of custom 3D printed puzzle boxes including HEX and a special Easter egg box with intricate details that function as mechanical moving parts